

# Python Full Stack Development Internship

Capstone Project: Markdown to HTML Converter

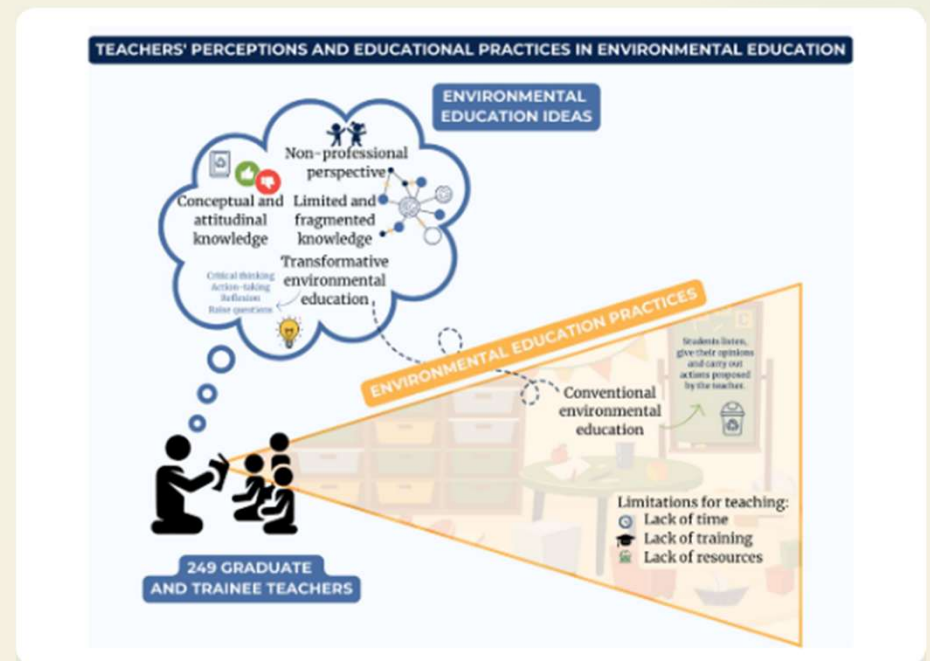
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**IILM University, Greater Noida**

# Organization & Internship Overview

- **Organization:** EduSkills Academy & AICTE.
- **Domain:** EdTech & Skill Development focusing on industry-relevant skills.
- **Duration:** 8-Week Virtual Internship (June 2026 - August 2026).
- **Objective:** Bridge the gap between academic knowledge and real-world application through hands-on full stack development.



# 8-Week Learning Journey

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## Weeks 3-4

React fundamentals, Hooks, state management, and API integration.

## Weeks 7-8

Testing, deployment strategies, and Capstone Project development.

## Weeks 1-2

Python Fundamentals, OOP, and FastAPI backend setup with Pydantic validation.

## Weeks 5-6

SQLAlchemy DB integration, advanced React patterns, and routing.



# Capstone Project

Markdown to HTML Converter

# Problem Statement & Objectives

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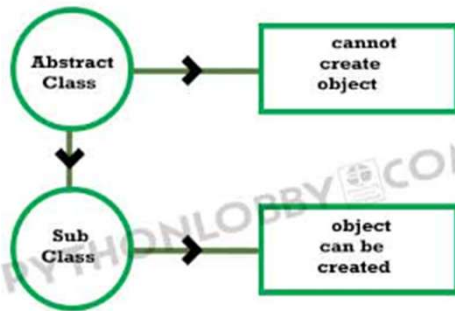
## The Problem

There is a constant need for utilities to automate the translation of plain text (Markdown) into equivalent HTML code for web display. This requires a robust parser that handles headers, emphasis, inline code, and hyperlinks seamlessly.

## Project Scope

Develop a Python-based sequential processing pipeline utilizing regular expressions to efficiently parse standard Markdown syntax and output valid, semantically correct HTML without unintended tag overlaps.

# Core Technologies Utilized



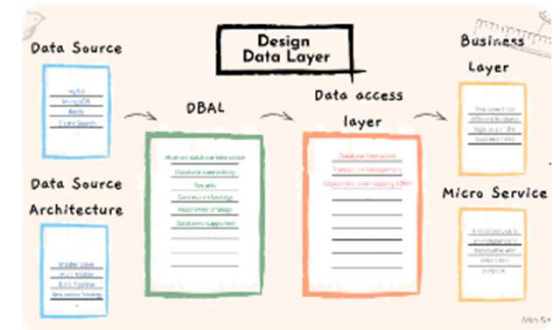
## Python Backend

String manipulation, regex (`re` module), and logic formulation.



## React & FastAPI

Full stack knowledge applied for robust API design and UI integration.

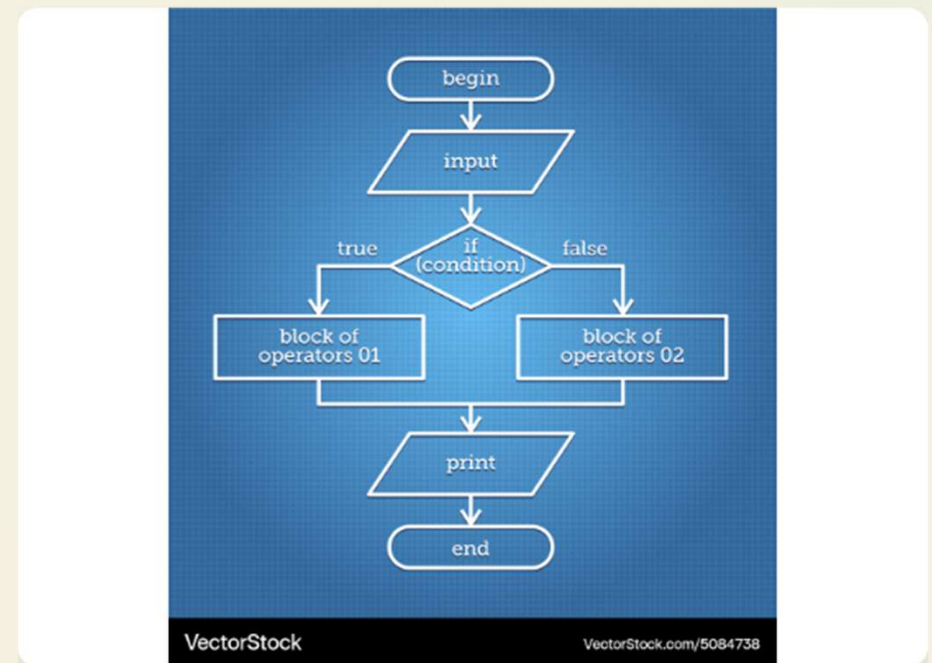


## SQLAlchemy

Relational database integration and ORM usage during the learning modules.

# Methodology & Pipeline

- **Header Parsing:** Identification of hash `#` prefixes to generate `h1` to `h6` HTML tags.
- **Emphasis Logic:** Regex non-greedy matching to convert asterisks to `strong` and `em` elements.
- **Inline Code:** Capturing text between backticks to wrap in `code` tags.
- **Hyperlinks:** Using capture groups ``\[((^\\)]+\\)\[((^)]+\\)`` to extract text and URLs.
- **Sequential Flow:** Processing in a specific order (Links > Bold > Italic > Code) to prevent formatting conflicts.



# Challenges & Solutions

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**Overlapping Regex Matches** Accurately handling nested Markdown syntax without unintended substitutions.

*Solution: Crafted specific regex patterns using non-greedy matching.*



**Order of Operations** Parsing italic text before bold text caused bold markers to break. *Solution: Enforced a strict sequential pipeline processing Links, then Bold, then Italics.*



**Semantic Validation** Ensuring the generated HTML is valid across different browser renderers. *Solution: Rigorous testing against expected HTML structures using unit tests.*

# Key Internship Outcomes

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**Weeks of Training**

## Hands-On Experience

Successfully transitioned from core Python fundamentals to advanced state management in React and database integration. The Capstone Project solidified problem-solving skills, regex proficiency, and the ability to build functional, sequential data pipelines from scratch.

# Thank You

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